

UNIT 7 · WEEKS 10–12

Threads and concurrency

📅 Weeks 10–12 · dates to be confirmed </> Lab TD6 (solved + AI challenge) ~ 5 animations

LAB TD6 · SOLVED

Threads and concurrency

Several threads share the memory of a process: it is fast and convenient, and also the source of the hardest bugs in programming. This lab causes them on purpose and fixes them with mutexes and semaphores.

✓ **Assignment: lab report**

Hand in this lab with a short LaTeX report (just what is needed). What it must cover, the template, an example and the Prism guide are in the Assignment tab. Generative AI allowed, with a declaration.

EXERCISES

- | | | | |
|---|--|---|--|
| 1 | Creating and joining threads | 4 | The dining philosophers |
| 2 | Ticket sellers: the race and two solutions | 5 | Exploration: thread cost, Peterson on an x86 and the futex (new) |
| 3 | Producers and consumers | | |

TD6 · 1

● Solved · tested

Creating and joining threads

Statement. Create 8 threads with `pthread_create` that run `start` and print the process pid (passed as an argument). The main thread waits for all of them with `pthread_join`. Add a global and a local variable to `start`, increment them and print their addresses and values.

Solution



C tds/td6/threads.c

```

/*
 * TD6 - exercise 1: create 8 threads, wait for them, and compare a global
 and a local variable.
 * All threads share the global `shared` (same address) but each has its own
 `local` (on its
 * own stack): the addresses printed show it. `shared++` is a race: see
 exercise 2.
 * Build: gcc -pthread (the Makefile already does it).
 */
#include <pthread.h>
#include <stdint.h>
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

#include "utils.h"

#define NTHREADS 8

static int shared = 0;

static void *start(void *arg)
{
    pid_t pid = (pid_t)(intptr_t)arg;      /* the pid travels inside the
void * */
    int local = 0;

    shared++;
    local++;
    printf("thread %lu of process %d: shared[%p] = %d, local[%p] = %d\n",
        (unsigned long)pthread_self() % 100000, pid, (void *)&shared,
shared, (void *)&local, local);
    return NULL;                          /* same as pthread_exit(NULL) */
}

int main(void)
{
    pthread_t tids[NTHREADS];
    pid_t pid = getpid();

    for (int i = 0; i < NTHREADS; i++) {
        int rc = pthread_create(&tids[i], NULL, start, (void *)
(intptr_t)pid);

```

```

        exit_if(rc != 0, "pthread_create"); /* pthread functions return the
error code */
    }
    for (int i = 0; i < NTHREADS; i++)
        exit_if(pthread_join(tids[i], NULL) != 0, "pthread_join");

    printf("process %d: all %d threads finished, shared = %d\n", pid,
NTHREADS, shared);
    return EXIT_SUCCESS;
}

```

Real output

```

$ ./threads
thread 57632 of process 357789: shared[0x650ff476602c] = 1,
local[0x7626e31fee74] = 1
thread 79520 of process 357789: shared[0x650ff476602c] = 4,
local[0x7626e19fbe74] = 1
thread 19968 of process 357789: shared[0x650ff476602c] = 5,
local[0x7626dbffee74] = 1
thread 64928 of process 357789: shared[0x650ff476602c] = 2,
local[0x7626e29fde74] = 1
thread 72224 of process 357789: shared[0x650ff476602c] = 3,
local[0x7626e21fce74] = 1
thread 86816 of process 357789: shared[0x650ff476602c] = 6,
local[0x7626e11fae74] = 1
thread 94112 of process 357789: shared[0x650ff476602c] = 7,
local[0x7626e09f9e74] = 1
thread 27264 of process 357789: shared[0x650ff476602c] = 8,
local[0x7626db7fde74] = 1
process 357789: all 8 threads finished, shared = 8

```

All threads print the **same pid** (they are one process) and the **same address** for `shared`, but every `local` sits at a different address: each thread has its own stack. The value of `shared` comes out of order and could even repeat: `shared++` is not atomic.

Ticket sellers: the race and two solutions

Statement. For a faculty party, 8 students (threads) sell in parallel the tickets of a global counter: while any remain, they call `sched_yield`, subtract one and count their sale; at the end they return how many they sold with `pthread_exit` and the main thread adds them up. Try growing amounts. What happens? Where is the critical section? Protect it with a semaphore and with a mutex.

Solution

C tds/td6/tickets_race.c

```
static volatile int tickets = 1000;    /* volatile only prevents caching in a
register:

                                         it does NOT make the accesses
atomic */

static void *seller(void *arg)
{
    int id = (int)(intptr_t)arg;
    long sold = 0;

    while (tickets > 0) {               /* critical section starts here ...
*/
        sold++;
        sched_yield();                 /* do not touch this line
*/
        tickets--;                     /* ... and ends here
*/
    }
    printf("[%d] I sold %ld tickets\n", id, sold);
    return (void *)(intptr_t)sold;     /* same as pthread_exit((void *)sold)
*/
}
```

Real output

```
$ ./tickets_race 100000 | tail -3
[5] I sold 14026 tickets
[7] I sold 14753 tickets
103195 tickets sold out of 100000 (tickets = -7)
```

More tickets than available are sold and the counter ends negative. The critical section is the whole span from `tickets > 0` to `tickets--`: between the test and the subtraction another thread can sell the last ticket, and `tickets--` is itself read-subtract-write, three steps that can interleave.

C tds/td6/tickets_mutex.c

```
static int tickets = 10000;
static pthread_mutex_t lock = PTHREAD_MUTEX_INITIALIZER;

static void *seller(void *arg)
{
    int id = (int)(intptr_t)arg;
    long sold = 0;

    for (;;) {
        pthread_mutex_lock(&lock);
        if (tickets ≤ 0) {                                /* checked with the lock held */
            pthread_mutex_unlock(&lock);
            break;
        }
        sold++;
        sched_yield();
        tickets--;
        pthread_mutex_unlock(&lock);
    }
    printf("[%d] I sold %ld tickets\n", id, sold);
    return (void *) (intptr_t)sold;
}
```

Real output

```
$ ./tickets_mutex 100000 | tail -3
[3] I sold 11577 tickets
[2] I sold 11338 tickets
100000 tickets sold out of 100000 (tickets = 0)
```

C tds/td6/tickets_sem.c

```
static sem_t available;

static void *seller(void *arg)
{
    int id = (int)(intptr_t)arg;
    long sold = 0;

    while (sem_trywait(&available) == 0) {    /* took one ticket */
        sold++;
        sched_yield();
    }
    exit_if(errno != EAGAIN, "sem_trywait"); /* EAGAIN = no tickets left,
    anything else is a bug */
    printf("[%d] I sold %ld tickets\n", id, sold);
    return (void *)(intptr_t)sold;
}
```

Real output

```
$ ./tickets_sem 100000 | tail -3
[6] I sold 14469 tickets
[2] I sold 11537 tickets
100000 tickets sold out of 100000 (tickets = 0)
```

With the **mutex**, the test and the subtraction happen with the lock held (and the test is repeated inside: the value read before taking it may be stale). With the **semaphore** the design is even better: the semaphore *is* the ticket counter, and `sem_trywait` "checks and takes" in one indivisible operation.



Fixed

The inherited mutex solution did not compile (`nbtickets_venduss`) and read the counter outside the lock.

Animation A7.1 · Race condition →

Producers and consumers

Statement. (a) A producer adds 1 to every cell of an array **A** and a consumer adds up all its cells, **LOOP** times each. Make them two threads that **take turns**, each with exclusive access to the whole array. (b) Now there is 1 producer and 4 consumers: every produced value must be consumed exactly once. Which new critical sections appear?

Solution

C tds/td6/prodcons.c

```
/*
 * TD6 - exercise 3 (a): one producer and one consumer take turns on the
 * WHOLE array.
 * The producer adds 1 to every cell, then the consumer sums every cell; LOOP
 * rounds.
 * Expected sum: SIZE * (1 + 2 + ... + LOOP) = SIZE * LOOP * (LOOP + 1) / 2.
 *
 * Two semaphores implement the turn-taking ("baton passing"):
 * can_produce = 1 at start (the producer goes first), can_consume = 0.
 * The critical section is the traversal of the array: while one thread walks
 * it, the
 * other must not touch it. With the two semaphores, at most one of them is
 * inside.
 */
#include <pthread.h>
#include <semaphore.h>
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

#include "utils.h"

#define SIZE 10000
#define LOOP 10

static int A[SIZE];
static long sum = 0;
static sem_t can_produce, can_consume;

/* "Opaque" functions of the statement: they must not be modified. */
static int consume(int *value) { usleep(1); return *value; }
```

```
static void produce(int v, int *value) { usleep(1); *value += v; }

static void *producer(void *arg)
{
    (void)arg;
    for (int i = 0; i < LOOP; i++) {
        sem_wait(&can_produce);                /* my turn */
        for (int j = 0; j < SIZE; j++)
            produce(1, &A[j]);
        printf("producer: round %d written\n", i + 1);
        sem_post(&can_consume);                /* your turn */
    }
    return NULL;
}

static void *consumer(void *arg)
{
    (void)arg;
    for (int i = 0; i < LOOP; i++) {
        sem_wait(&can_consume);
        for (int j = 0; j < SIZE; j++)
            sum += consume(&A[j]);
        printf("consumer: sum after round %d = %ld\n", i + 1, sum);
        sem_post(&can_produce);
    }
    return NULL;
}

int main(void)
{
    pthread_t p, c;

    exit_if(sem_init(&can_produce, 0, 1) == -1 || sem_init(&can_consume, 0,
0) == -1, "sem_init");
    exit_if(pthread_create(&p, NULL, producer, NULL) != 0, "pthread_create");
    exit_if(pthread_create(&c, NULL, consumer, NULL) != 0, "pthread_create");
    exit_if(pthread_join(p, NULL) != 0 || pthread_join(c, NULL) != 0,
"pthread_join");

    long expected = (long)SIZE * LOOP * (LOOP + 1) / 2;
    printf("sum = %ld, expected %ld: %s\n", sum, expected, sum == expected ?
"OK" : "WRONG");
    sem_destroy(&can_produce);
    sem_destroy(&can_consume);
}
```

```
    return sum == expected ? EXIT_SUCCESS : EXIT_FAILURE;
}
```

Real output

```
$ ./prodcons | tail -4
consumer: sum after round 9 = 450000
producer: round 10 written
consumer: sum after round 10 = 550000
sum = 550000, expected 550000: OK
```

(a) Two semaphores pass the turn like a relay baton: `can_produce` starts at 1 and `can_consume` at 0. Each thread waits for its semaphore, walks the array and enables the other. The expected sum is $\text{SIZE} \times (1 + 2 + \dots + \text{LOOP}) = 10,000 \times 55$.

C tds/td6/prodcons_multi.c

```
/*
 * TD6 - exercise 3 (b): one producer, N consumers, and a bounded buffer (the
 * classic of the
 * lecture). The producer puts ITEMS values in a circular buffer of BUF
 * cells; the consumers
 * take them out and add them to a shared sum. Every value must be consumed
 * exactly once.
 *
 *
 * empty = number of free cells (starts at BUF) - the producer waits on
 * it
 * full  = number of used cells (starts at 0) - the consumers wait on
 * it
 * mutex = protects the `out` index shared by the consumers (new critical
 * section!)
 *
 * sum_lock protects the shared sum
 * To stop, the producer sends one "poison pill" (value -1) per consumer.
 * ./prodcons_multi [consumers] [items] (default 4 consumers, 100000
 * items)
 */
#include <pthread.h>
#include <semaphore.h>
#include <stdint.h>
#include <stdio.h>
#include <stdlib.h>

#include "utils.h"

#define BUF 16
```

```
#define POISON (-1)

static int buffer[BUF];
static int in = 0, out = 0;           /* in: producer only; out: shared
by consumers */
static sem_t empty, full;
static pthread_mutex_t out_lock = PTHREAD_MUTEX_INITIALIZER;
static pthread_mutex_t sum_lock = PTHREAD_MUTEX_INITIALIZER;
static long long sum = 0;
static int nconsumers = 4, items = 100000;

static void *producer(void *arg)
{
    (void)arg;
    for (int v = 1; v ≤ items + nconsumers; v++) {
        int value = v ≤ items ? v : POISON;
        sem_wait(&empty);           /* wait for a free cell */
        buffer[in] = value;         /* only one producer: no lock
needed for `in` */
        in = (in + 1) % BUF;
        sem_post(&full);           /* one more item available */
    }
    return NULL;
}

static void *consumer(void *arg)
{
    long mine = 0, count = 0;
    for (;;) {
        sem_wait(&full);
        pthread_mutex_lock(&out_lock); /* two consumers must not take
the same cell */
        int value = buffer[out];
        out = (out + 1) % BUF;
        pthread_mutex_unlock(&out_lock);
        sem_post(&empty);
        if (value == POISON)
            break;
        mine += value;
        count++;
    }
    pthread_mutex_lock(&sum_lock); /* add the partial sum once, not
per item */
    sum += mine;
    pthread_mutex_unlock(&sum_lock);
}
```

```

    printf("consumer %d: %ld items\n", (int)(intptr_t)arg, count);
    return NULL;
}

int main(int argc, char *argv[])
{
    if (argc > 1) nconsumers = atoi(argv[1]);
    if (argc > 2) items = atoi(argv[2]);
    pthread_t prod, cons[nconsumers];

    exit_if(sem_init(&empty, 0, BUF) == -1 || sem_init(&full, 0, 0) == -1,
"sem_init");
    exit_if(pthread_create(&prod, NULL, producer, NULL) != 0,
"pthread_create");
    for (int i = 0; i < nconsumers; i++)
        exit_if(pthread_create(&cons[i], NULL, consumer, (void *) (intptr_t)i)
!= 0, "pthread_create");
    exit_if(pthread_join(prod, NULL) != 0, "pthread_join");
    for (int i = 0; i < nconsumers; i++)
        exit_if(pthread_join(cons[i], NULL) != 0, "pthread_join");

    long long expected = (long long)items * (items + 1) / 2;
    printf("sum = %lld, expected %lld: %s\n", sum, expected, sum == expected
? "OK" : "WRONG");
    return sum == expected ? EXIT_SUCCESS : EXIT_FAILURE;
}

```

Real output

```

$ ./prodcons_multi 4 100000
consumer 2: 23918 items
consumer 3: 26423 items
consumer 1: 23037 items
consumer 0: 26622 items
sum = 5000050000, expected 5000050000: OK

```

(b) The classic bounded-buffer problem: `empty` counts free cells, `full` used cells. With several consumers a new critical section appears: the `out` index, protected by a mutex (otherwise two consumers would take the same cell). Each consumer's sum is added once, under another mutex. To finish, the producer sends one "poison pill" (-1) per consumer.



Consistency fixed

The statement talked about `producteur / consommateur` and the inherited solutions about `lecteur / ecrivain`; one of them did not meet "exclusive access to the whole array" and the multi-consumer one busy-waited with `sem_trywait + usleep`. Statement, code and names now match.

TD6 · 4

● Solved · tested (new)

The dining philosophers

Statement. 8 philosophers sit at a round table with a chopstick between each pair; to eat, each needs the chopstick on the right and the one on the left (one mutex per chopstick). Everybody takes the right one first, then the left one, and repeats think–take–eat–release. What happens? Propose a change that avoids it.

Solution

C tds/td6/philosophers.c

```
/*
 * TD6 - exercise 4: the dining philosophers. N philosophers, N chopsticks
 (one mutex each).
 * Every philosopher takes the RIGHT chopstick, then the LEFT one.
 * DEMO_DEADLOCK (Makefile target philosophers_deadlock): everybody may
 hold the right
 * chopstick and wait forever for the left one: circular wait = deadlock. A
 pause between
 * the two lock operations makes it happen almost every time.
 * default (solution): break the circular wait by ordering the resources -
 every
 * philosopher takes the chopstick with the SMALLER number first. The last
 philosopher
 * (whose right chopstick is N-1 and left is 0) therefore starts with the
 left one.
 * ./philosophers [meals] (default 5 meals each)
 */
#include <pthread.h>
#include <stdint.h>
#include <stdio.h>
```

```

#include <stdlib.h>
#include <unistd.h>

#include "utils.h"

#define N 8

static pthread_mutex_t chopstick[N];
static int meals = 5;

static void think(int id)
{
    printf("%2d: thinking\n", id);
    usleep((useconds_t)(random() % 2000));
    printf("%2d: hungry\n", id);
}

static void eat(int id)
{
    printf("%2d: eating\n", id);
    usleep((useconds_t)(random() % 2000));
    printf("%2d: full\n", id);
}

static void *philosopher(void *arg)
{
    int id = (int)(intptr_t)arg;
    int right = id, left = (id + 1) % N;
#ifdef DEMO_DEADLOCK
    int first = right, second = left;
#else
    int first = right < left ? right : left;      /* global order on the
chopsticks */
    int second = right < left ? left : right;
#endif
    for (int m = 0; m < meals; m++) {
        think(id);
        pthread_mutex_lock(&chopstick[first]);
#ifdef DEMO_DEADLOCK
        usleep(20000);                          /* everybody grabs the
first one... */
#endif
        pthread_mutex_lock(&chopstick[second]);
        eat(id);
        pthread_mutex_unlock(&chopstick[second]);
    }
}

```

```

        pthread_mutex_unlock(&chopstick[first]);
    }
    printf("%2d: done after %d meals\n", id, meals);
    return NULL;
}

int main(int argc, char *argv[])
{
    pthread_t tids[N];

    if (argc > 1)
        meals = atoi(argv[1]);
    setvbuf(stdout, NULL, _IOLBF, 0);    /* line by line, even into a pipe:
see who is stuck */
    srand(2026);
    for (int i = 0; i < N; i++)
        exit_if(pthread_mutex_init(&chopstick[i], NULL) != 0,
"pthread_mutex_init");
    for (int i = 0; i < N; i++)
        exit_if(pthread_create(&tids[i], NULL, philosopher, (void *)
(intptr_t)i) != 0, "pthread_create");
    for (int i = 0; i < N; i++)
        exit_if(pthread_join(tids[i], NULL) != 0, "pthread_join");
    printf("dinner finished: %d philosophers x %d meals\n", N, meals);
    return EXIT_SUCCESS;
}

```

Real output

```

$ timeout 3 ./philosophers_deadlock 3 | tail -8; echo "exit status:
${PIPESTATUS[0]} (124 = still blocked after 3 s)"
7: hungry
2: hungry
3: hungry
1: hungry
0: hungry
6: hungry
4: hungry
5: hungry
exit status: 124 (124 = still blocked after 3 s)

```

Real output

```
$ ./philosophers 3 | tail -3
7: full
7: done after 3 meals
dinner finished: 8 philosophers x 3 meals
```

With everybody taking the right one first, it is enough that all 8 do it at the same time for each one to wait for the chopstick its neighbour holds: **deadlock**. The real output shows all 8 "hungry" and the program stuck until `timeout` kills it. The four Coffman conditions hold (mutual exclusion, hold and wait, no preemption and circular wait). The solution breaks the **circular wait**: chopsticks are ordered and each philosopher takes the lower-numbered one first; the last philosopher therefore takes the left one first and the cycle cannot close.

[Animation A7.3 · Deadlock →](#)

TD6 · 5

● New · solved · tested

Exploration: thread cost, Peterson on an x86 and the futex

Statement. Three measurements to check (or refute) three claims of the slides: (1) `create_cost`: is creating a thread cheaper than creating a process? (slide 136). (2) `peterson`: Peterson's algorithm "is correct" (slide 152)... on a real processor? Compare with `peterson_fence`. (3) `lock_cost` under `strace -c`: how many system calls does a mutex make with and without contention? (slides 160–162).

Solution and real outputs

C tds/td6/create_cost.c

```
/*
 * TD6 - exploration 1: how much does creating a thread cost compared with a
 * process? (original slide 136)
 * Creates and waits for N threads (pthread_create + pthread_join) and N
 * processes (fork + waitpid, the
 * child exits at once) and prints the average time of each.
 * ./create_cost [N]      default: 2000
 */
#include <pthread.h>
```

```

#include <stdio.h>
#include <stdlib.h>
#include <sys/wait.h>
#include <time.h>
#include <unistd.h>

#include "utils.h"

static double now(void)
{
    struct timespec ts;
    exit_if(clock_gettime(CLOCK_MONOTONIC, &ts) == -1, "clock_gettime");
    return ts.tv_sec + ts.tv_nsec / 1e9;
}

static void *nothing(void *arg)
{
    return arg;
}

int main(int argc, char *argv[])
{
    int n = argc > 1 ? atoi(argv[1]) : 2000;

    double t0 = now();
    for (int i = 0; i < n; i++) {
        pthread_t t;
        int rc = pthread_create(&t, NULL, nothing, NULL);
        exit_if(rc != 0, "pthread_create");
        exit_if(pthread_join(t, NULL) != 0, "pthread_join");
    }
    double thread_us = (now() - t0) * 1e6 / n;

    t0 = now();
    for (int i = 0; i < n; i++) {
        pid_t pid = fork();
        exit_if(pid == -1, "fork");
        if (pid == 0)
            _exit(0);
        exit_if(waitpid(pid, NULL, 0) == -1, "waitpid");
    }
    double fork_us = (now() - t0) * 1e6 / n;

    printf("%d threads:   %.1f us per pthread_create + pthread_join\n", n,
        thread_us);
}

```

```
printf("%d processes: %.1f us per fork + waitpid\n", n, fork_us);
printf("a process costs %.1f times a thread\n", fork_us / thread_us);
return EXIT_SUCCESS;
}
```

Real output

```
$ ./create_cost 2000
2000 threads:      41.2 us per pthread_create + pthread_join
2000 processes:   248.8 us per fork + waitpid
a process costs 6.0 times a thread
```

A thread copies no page tables or descriptors: it only reserves a stack and a task structure. In Linux both are created with `clone`; what changes is what is shared.

C tds/td6/peterson.c

```
/*
 * TD6 - exploration 2: Peterson's algorithm on a real multicore x86
 (original slides 152 and 154).
 * Two threads increment a shared counter N times each, protecting the
 increment with Peterson's lock.
 * The algorithm is correct on paper, but it "assumes memory accesses are
 sequentially ordered per thread"
 * (slide 154). Without a fence, the x86 store buffer lets each thread read
 the other's flag BEFORE its own
 * store is visible: both enter the critical section and increments are lost.
 With a full fence
 * (mfence, via __atomic_thread_fence) the algorithm works. Compiled twice:
 peterson (no fence) and
 * peterson_fence (-DFENCE).
 * ./peterson [N]      default: 2000000
 */
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>

#include "utils.h"

static volatile int flag[2];
static volatile int turn;
static volatile long counter;
static long n;

static void *worker(void *arg)
```

```
{
    int me = (int)(long)arg, other = 1 - me;
    for (long i = 0; i < n; i++) {
        flag[me] = 1;
        turn = other;
#ifdef FENCE
        __atomic_thread_fence(__ATOMIC_SEQ_CST);    /* the stores must be
visible before the loads */
#endif
        while (flag[other] && turn == other)
            ;    /* busy waiting */
        counter = counter + 1;    /* critical section */
        flag[me] = 0;
    }
    return NULL;
}

int main(int argc, char *argv[])
{
    n = argc > 1 ? atol(argv[1]) : 2000000;
    pthread_t t[2];
    for (long i = 0; i < 2; i++)
        exit_if(pthread_create(&t[i], NULL, worker, (void *)i) != 0,
"pthread_create");
    for (int i = 0; i < 2; i++)
        exit_if(pthread_join(t[i], NULL) != 0, "pthread_join");
#ifdef FENCE
    const char *variant = "with fence";
#else
    const char *variant = "no fence";
#endif
    printf("Peterson %-10s: counter = %ld, expected %ld, lost %ld\n",
variant, counter, 2 * n, 2 * n - counter);
    return EXIT_SUCCESS;
}
```

Real output

```
$ for i in 1 2 3 4 5; do ./peterson; done; for i in 1 2 3 4 5; do
./peterson_fence; done
Peterson no fence : counter = 4000000, expected 4000000, lost 0
Peterson no fence : counter = 4000000, expected 4000000, lost 0
Peterson no fence : counter = 4000000, expected 4000000, lost 0
Peterson no fence : counter = 4000000, expected 4000000, lost 0
Peterson no fence : counter = 4000000, expected 4000000, lost 0
Peterson with fence: counter = 4000000, expected 4000000, lost 0
Peterson with fence: counter = 4000000, expected 4000000, lost 0
Peterson with fence: counter = 4000000, expected 4000000, lost 0
Peterson with fence: counter = 4000000, expected 4000000, lost 0
Peterson with fence: counter = 4000000, expected 4000000, lost 0
```

Without a fence, some runs lose increments: both threads were in the critical section at the same time. The x86 leaves each write in a **buffer** and lets a thread read the other's note before its own is visible (the TSO model). `__atomic_thread_fence` (an `mfence` instruction) prevents it. It is what slide 154 warns about and what animation A7.5 verifies, state by state.

C tds/td6/lock_cost.c

```
static void *worker(void *arg)
{
    for (long i = 0; i < n; i++) {
        pthread_mutex_lock(&lock);
        counter++;
        pthread_mutex_unlock(&lock);
    }
    return arg;
}
```

Real output

```
$ ./lock_cost 1; ./lock_cost 8
1 thread(s) x 1000000 lock/unlock: counter = 1000000, 28.2 ns per lock/unlock
8 thread(s) x 1000000 lock/unlock: counter = 8000000, 184.2 ns per
lock/unlock
```

Real output

```
$ strace -f -c -e trace=futex ./lock_cost 1 100000; strace -f -c -e
trace=futex ./lock_cost 8 100000
1 thread(s) x 100000 lock/unlock: counter = 100000, 48.4 ns per lock/unlock
0.00    0.000000    0    1    futex
8 thread(s) x 100000 lock/unlock: counter = 800000, 85.5 ns per lock/unlock
100.00  0.223730    27    8013    3116 futex
```

Without contention, 100,000 `lock` / `unlock` make **no** `futex` call (the only one is the `join`'s): a free mutex is taken with an atomic instruction in user space. Only when a thread must wait does it enter the kernel. It is the "limited busy waiting" of slide 160, made real by the Linux *futex*.

See it verified → A7.5 the milk problem

Checklist

Generative AI challenge · Philosophers: detect the deadlock with a graph

Goal. Instrument ``philosophers.c`` to record who waits for whom and detect the cycle of the wait-for graph automatically; compare two solutions (global order and a waiter semaphore).

Starting prompt. Use it in the tool you prefer (Claude, ChatGPT, Gemini, Copilot, Codex...) with the *solved* version of the lab as context. Ask for a plan first and review every step before accepting it.

Prompt

I have the dining philosophers with mutexes [paste philosophers.c]. Help me: (1) add a watchdog thread that every 100 ms builds the graph "philosopher i waits for the chopstick held by j" and detects a cycle; (2) implement the waiter solution (a semaphore that lets N-1 philosophers sit down) and compare it with the global-order one in meals per second; (3) link each solution with the Coffman condition it breaks. Before coding, ask me which of the 4 conditions each one breaks.

How to check the result

- The watchdog detects the cycle in ``philosophers_deadlock`` and prints the 8 philosophers of the cycle.
- Both solutions always finish (repeat 20 times).
- Your meals-per-second numbers come from a measurement, not from the AI.

Declare your use of AI: tool, relevant prompts, what you checked yourself and what you fixed. What counts is that you can explain and verify every decision.